Area Science Park
Research and Technology Institute
Laboratory of Data Engineering

Web Application for DPCfam and DPCstruct Data Exploration

Report 2 (19 Jan - 26 Jan 2026) : Web Design (Frontend)

Emmanuel Nyandu Kagarabi
Master in Data Management and Curation (MDMC - SISSA)

Supervisors :
Valerio Piomponi
Elaheh Saadat

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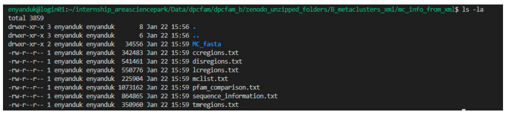
1 Introduction

This report is the second of a sequence in which we describe the accomplishment of timely (weekly) assigned tasks during our internship. The main focus here is the Web design(Frontend) side of our browser-based web application. That app should allow users to navigate with ease the datasets described in our previous report, namely DPCFam(Russo et al., 2022) and DPCStruct(Barone et al., 2025).

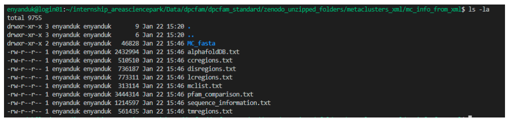
2 Development

2.1 Report 1 Appendix

Before jumping into the topic, in the previous report, for the DPCFam Dataset in particular, we conducted our experiments on a small dataset(`DPCFamB`) (Figure 1a) which is representative but incomplete (Figure 1).



(a) DPCFamB



(b) `DPCFam Standard dataset`

Figure 1: DPCFam datasets. Both datasets have almost the same structure, except one additional file in the standard dataset. It is worth noting that these datasets are completely disjoint(based on MC_IDs) but complementary, somehow.

Thanks to Orfeo, we explored the standard database, running the `parse.sh` file as previously, we obtained the same list of files as `DPCFam_B`, but with an additional file `alphaFoldDB.txt` (Figure 2) :

	#AF_protein	MC_ID	MC_hmm_length	e-value	AF_seq_start	AF_seq_end	MC_hmm_start	MC_hmm_end	MC_hmm_coverage	average_plddt
0	AF-P0A149-F1	MC1	119	2.800000e-23	4	181	1	112	0.941176	96.2441
1	AF-Q04671-F1	MC4	63	1.300000e-29	770	830	2	62	0.968254	93.7970
2	AF-A0A1C1CP96-F1	MC19	71	1.300000e-40	66	135	2	71	0.985915	83.6246
3	AF-P97089-F1	MC21	93	1.500000e-35	1	93	1	93	1.000000	93.5686
4	AF-P9WJB2-F1	MC24	90	4.500000e-26	161	250	2	89	0.977778	89.7296

Figure 2: Additional file in the Standard DPCFam dataset.

For each MetaCluster(`MC_ID`), a representative AlphaFoldDB structure(Figure 2) is reported, along with the following:

- E-value : `e-value` column.

- Sequence range : `AF_seq_start` and `AF_seq_end` columns.
- hmm range : `MC_hmm_start` and `MC_hmm_end` columns.
- hmm coverage : `MC_hmm_coverage` column.
- plddt[1] value : `average_plddt` column.

For the rest of this report, we will suggest a first draft of our web app for the Exploration of DPCFam(prioritising `DPCFam Standard` which is complete) and DPCStruct.

It is worth noting as well that DPCStruct's input is the output of Foldseek. In turn, Foldseek outputs clusters from AlphaFoldDB. AlphaFoldDB outputs predictions of structures for a majority of proteins from UniProtKB. For each prediction, AlphaFoldDB also outputs the pLDDT, which is a similarity score between the predicted structure and the true structure of a protein; therefore, the higher the pLDDT, the better.

The main outputs of DPCStruct (MCs) were compared to some existing powerful (and reliable) classification tools such as PFam, CATH, and SCOP. Efforts are still ongoing on our side to understand and/or point out some relevant connections (possibly in upcoming reports).

[1] **Predicted Local Distance Difference Test (pLDDT)** is a per-residue confidence metric (0–100) used by AlphaFold and other structure prediction tools to estimate the accuracy of a 3D protein model. Higher scores indicate higher confidence in the local atomic positions. Generally, pLDDT > 90 indicates high confidence, while < 50 suggests low confidence.

2.2 Frontend side of our Django Web App

2.2.1 Landing Page (Home)

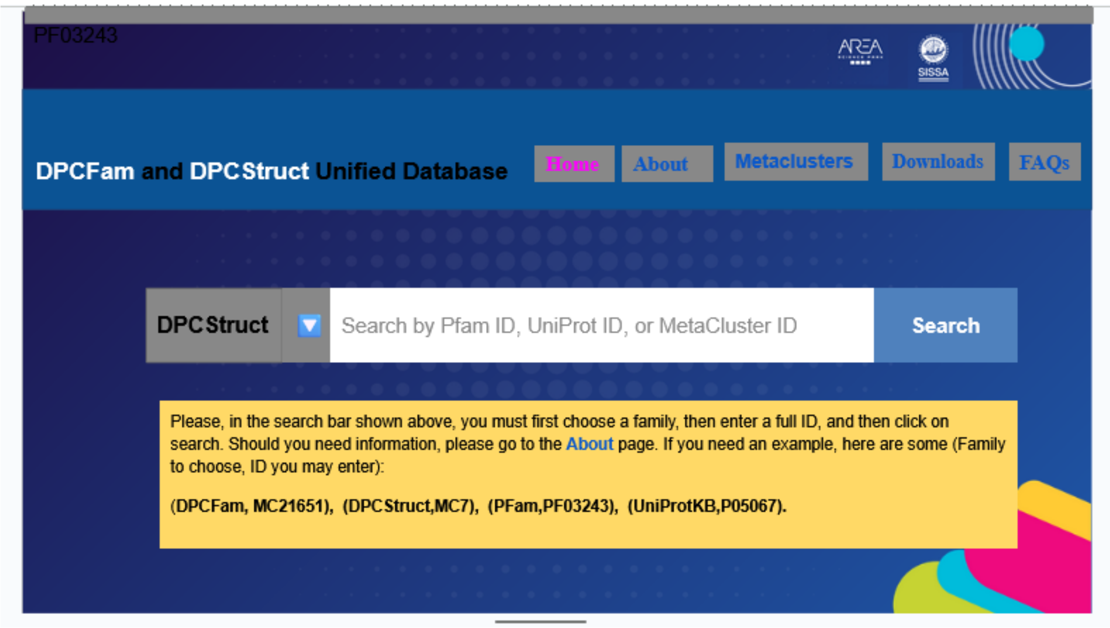


Figure 3: Home Page : Four choices for any user: DPCFam, DPCStruct, PFam and UniProtKB, and then entering a valid ID in the global search bar.

According to this first draft, once a user arrives at the landing page (Figure 3, they may read a short description (approximately 2 lines). If interested, they can scroll down to the global search bar, where they must make a unique choice of family among the four possible ones (**DPCFam**,**DPCStruct**,**PFam** and **UniProtKB**) followed by typing a valid ID according to the chosen family and then, submitting the query. What to expect at that step?, four conditional options arise:

Option 1 : The user has chosen DPCFam with a valid Metacluster ID

1. In case the MCID doesn't exit, we can display a short response (*There's no info related to your choice, please enter a different ID.*)

2. If everything goes well (the MC ID is valid and exists in our database (MC100, for instance), we can nagivate through tables summarizing the query:

MC - Metacluster Summary

Summary | Sequences | Cross-links | Downloads

Tab 1 : Summary

Table 1: DPCFam, MC100 : Properties.

Number of sequences (UniRef50):	81
Average sequence length:	108±13 aa
Average transmembrane regions:	0
Low complexity (%):	0.62
Coiled coils (%):	0
Disordered domains (%):	17.4
Pfam dominant architecture:	None
Pfam % dominant architecture:	0
Pfam overlap:	0
Pfam overlap type:	**None**
AlphafoldDB representative:	AF-X8FG90-F1 (24-133) - AlphafoldDB

Tab 2 : Sequences

Table 2: MC100 Sequences.

Protein	Range	AA
A0A0N0CXG0	392-505	KHTQKKNEEINNLRGHLKISDSILHKIGKQSVDIIPWDIANIYYNQLQYTPRPVMQSYAVQNKYLIDLNTKKYYSSNAPEFVFYRNSSIDDRYPYWDDVGVK
R9IHF1	19-110	LPEAISKEIGDASVSIYPWEISYCASNNLNYIPLATIQAYSTYTPYLDKISAAKFLNSDPPEYILLTLNTIDNRWPFIECPQTWEAIKNNYY
A0A1X1EX29	414-517	TVDIYSYDQSYLLASGNKWNPRPVLQSYSVYTPKLAELNKMHLLGDNRPDNVIFGVQPIDRRLPSLEDGASWPVLLSSYEPTTFTGDYLYLKHRSTPTPS
X8FG90	15-138	SAEARQRRLYDIPSRFVKTIGPDPVHIDPDETSAVWAYDFAWHPAPIFQTYSVYTPALDKLNRDTLAAGPQFVLSLRSATSPATGINGRLGVQENPLYSRAL
UPI000345A602	432-558	PARVGAQLSSLLNLKAFQSSLNEISAADLAQIKLPEKAIRLVGNKAIDIVPWEVLLAEANQLNWKPRPVFQSYSAYTRFLDSANYKSLSSAPRDYIFYNFTSI
B2FLV4	403-495	DLYPWDLSPLLASGASWKPRPVPQSYAAFSPPLLDANAEHLRSSPPTRIFFNINPIDQRYPSMEDGASWLTLLGGFRAQRIDAGYALLSRRDA
A0A1Q7YUY4	1-102	MRILEGGTVDVVMSETSLVYLEGLRYRPRPVIQSYAAYDAYLDQVNADKLQAPGAPDFILFHVHPGGDRYWFSEETRTRLAILQWYDDIGRFENFLVLKRR

Tab 3 : Cross-links

Table 3: Cross-links

DPCStruct	List | Links
PFam	List
UniProtKB	List

Tab 4 : Downloads

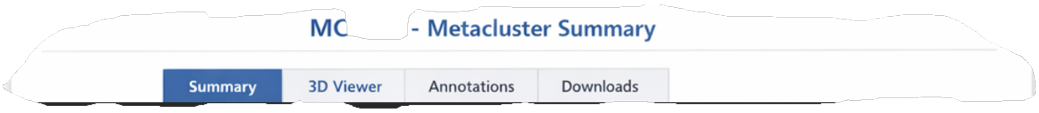
Table 4: Download MC100 Data

Seeds	MC100.fasta
MSA	MC100_msa.fasta
HMM Model	MC100.hmm

Option 2 : The user have chosen DPCStruct with a valid MC ID

1. In case the MCID doesn't exit, we can display a short response (*There's no info related to your choice, please enter a different ID.*)

2. If everything goes well (the MC ID is valid and exists in our database, e.g., MC1), we can display tables summarizing the query:



Tab 1 : Summary

Table 5: DPCStruct, MC1 : Properties.

mcID	49 953
size	28 992
len_aa	282.901
len_std	60.4128
len_ratio	0.213547
plddt	81.7385
disorder	0.225913
alntmscore	0.3476
tmscore	0.3795
lddt	0.3599
prob	0.6535
pident	110.5943

Tab 2 : 3D Viewer

Table 6: MC1 3D Viewer-Representative.

Representative	Range	3D Structure
A0A0A0BXF7	10 - 250	Visit me in AlphaFoldDB

Tab 3 : Annotations

Table 7: Annotations

DPCFam	List|Links
PFam	List of top 5
UniProtKB	List
CATH|SCOP	List of top matches with thresholds

Tab 4 : Downloads

Table 8: Download MC1 Representative Data

Seed	MC1.fasta
PDB	MC1.pdb

Option 3 : The user have chosen PFam with a valid ID

Whenever the ID is valid, we have to display all consistent samples in other families, resulting in the following:

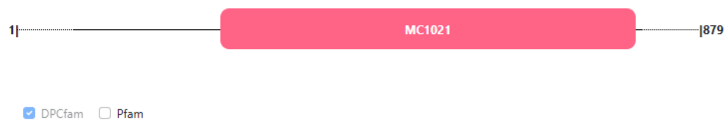
Table 9: PFam ID (PF10345) Matching Summary

DPCFam	DPCStruct
1. MC1021 88%	1. MC1021 88%
2. MC208863 37%	2. MC208863 37%
3. MC378188 32%	3. MC378188 32%
4. MC115382 27%	4. MC115382 27%

Option 4 : The user have chosen UniPRotKB with a valid ID

1. In case the ID doesn't exit, we can display a short response (*There's no info related to your choice, please enter another different ID.*)

2. If everything goes well (the ID is valid and exists somewhere in our database, we can display all matches as shown in Figure 4:



List of domains

DPCfam domains	Range	DPCfam web
MC1021	209-801	search

Pfam domains	Range	DPCfam web	Pfam web
PF10345	207-801	search	search

Figure 4: UniProt ID (U9UJD3) matches : We'll add a line of content(if any) related to DPCStruct MCs and the graphic displayed above may result into some overlapping fields.

2.2.2 About Page

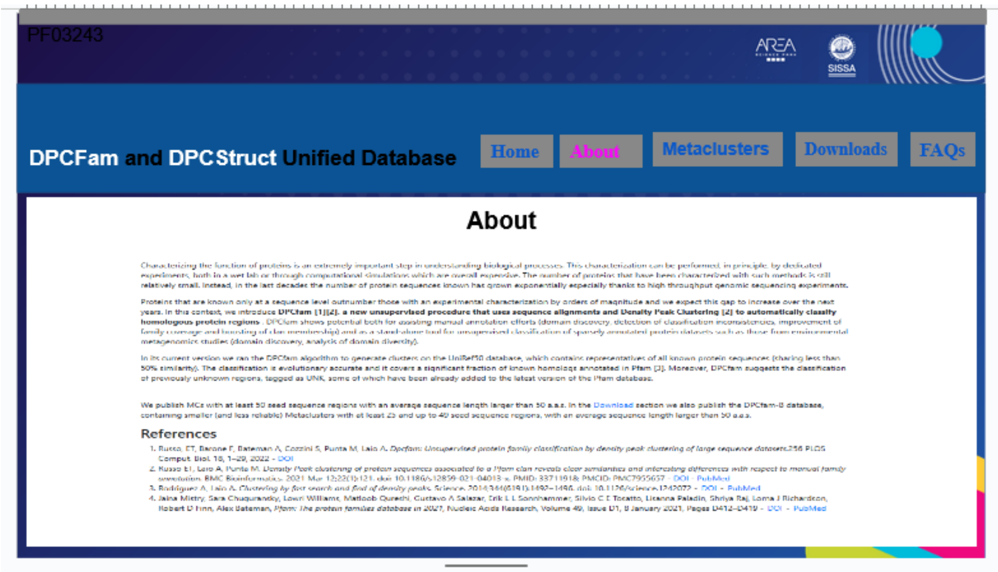


Figure 5: About Page : Useful info about the project.

2.2.3 Metaclusters Page

The user must choose between the two following families :

Figure 6: Metaclusters Page : Paginated tabs per family.

In case the user clicked on DPCFam, then we display all relevant properties related to all MCs:

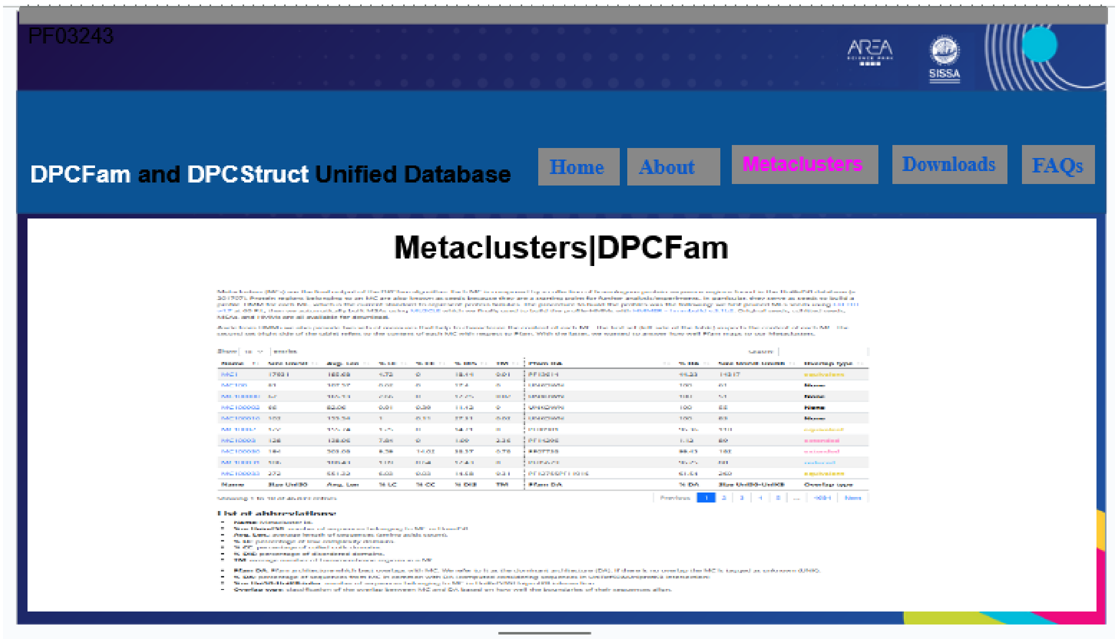


Figure 7: DPCFam Page : We present a paginated table (50-100 rows per page), we may add some additional features like : Sorting (based on size, avg length), Filtering(Consitent with Pfam or Not),...

- **Name** : Metacluster (MC) identifier.

- **Size UniRef50** : Number of sequences belonging to the MC in UniRef50.

- **Avg. Len.** : Average length of sequences (amino acid count).

- **% LC** : Percentage of low-complexity domains.

- **% CC** : Percentage of coiled-coil domains.

- **% DIS** : Percentage of disordered domains.

- **TM** : Average number of transmembrane regions in a metacluster.

- **Pfam DA** : Pfam architecture that best overlaps with the metacluster. This is referred to as the dominant architecture (DA). If no overlap exists, the metacluster is tagged as unknown (UNK).

- **% DA** : Percentage of sequences from the metacluster in common with the dominant architecture (computed considering sequences in the UniRef50 and UniProtKB intersection).

- **Size Uni50–UniKB inter** : Number of sequences belonging to the metacluster in the UniRef50–UniProtKB intersection.

- **Overlap type** : Classification of the overlap between the metacluster and the dominant architecture based on how well their sequence boundaries align.

In the case the user chose DPCStruct, then we can display properties related to MCs :

	mcID	size	len_aa	len_std	len_ratio	plddt	disorder	alntmscore	tmscore	lddt	prob	pident
0	61388	30643	210.005	38.0453	0.181164	82.6220	0.284676	0.2305	0.2478	0.3321	0.3082	10.7832
1	49953	28922	282.901	60.4128	0.213547	81.7385	0.225913	0.3476	0.3795	0.3599	0.6535	10.5943
2	52595	26312	122.438	21.1918	0.173082	84.0467	0.286809	0.3575	0.3819	0.3671	0.4309	10.8535
3	42223	17503	260.899	60.0432	0.230140	78.8912	0.223797	0.4164	0.4628	0.4638	0.7612	16.7721
4	124	16132	191.621	37.0314	0.193253	85.6541	0.197148	0.3631	0.3936	0.3936	0.5154	12.6375

Figure 8: MCs Properties

- `mcID`: Metacluster ID.
- `size`: Number of domains.
- `len_aa`: Average length of domains (number of amino acids).
- `len_std`: Standard deviation of domain lengths.
- `len_ratio`: Ratio of len_std to len_aa.
- `plddt`: Average predicted LDDT as reported by AlphaFold2.
- `disorder`: Average intrinsic disorder score calculated with AIUPred.
- `alntmscore`: Pairwise alignment TM-score between domains, averaged over all pairs.
- `tmscore`: Pairwise alignment TM-score between domains, averaged over all pairs, using the maximum between TM-score normalized by query or target.
- `lddt`: Pairwise LDDT score, averaged over all pairs.

- `prob`: Pairwise probability of homology according to SCOPe, as reported by Foldseek.

- `pident`: Pairwise percentage identity, averaged over all pairs.

In addition to all properties given above we can add: **MC Representative**,**PFam Consistency**, **CATH/SCOP Consistency**, resulting in a page like the following (Figure 9):

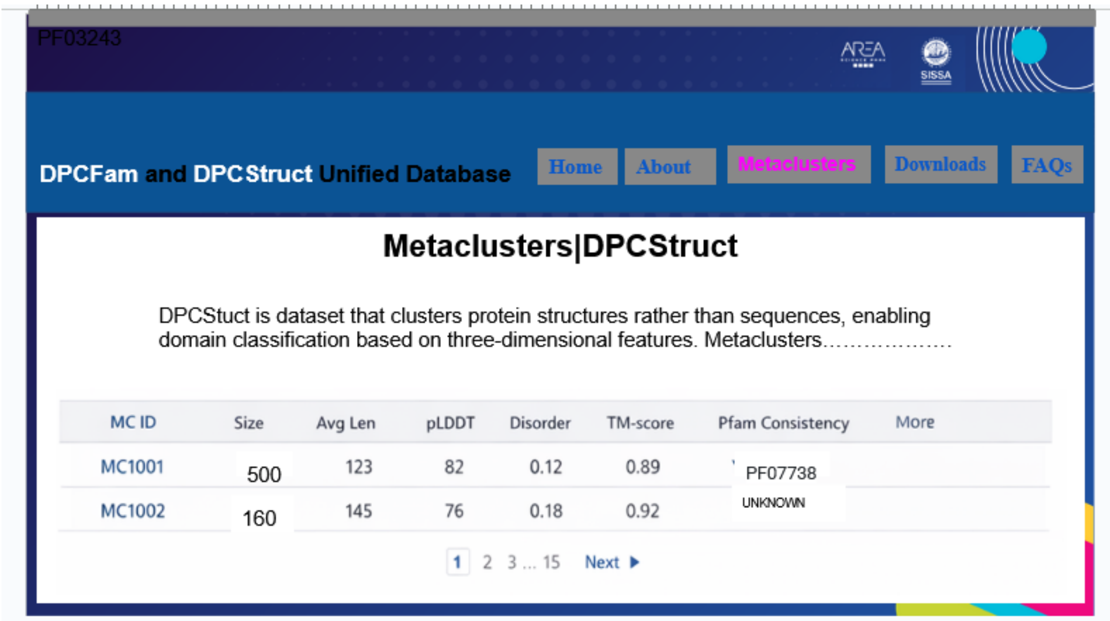


Figure 9: DPCStruct Page : We present a paginated table(50-100 rows per page), we may add some additional features like : Sorting (based on size, plDDT), Filtering,...

2.2.4 Downloads

The content to be downloaded depends on the chosen family (DPCFam, DPC-Struct):

Figure 10: Downloads Page

Case 1 : DPCFam

The user can download the following files :

Table 10: DPCFam files

File	Description	Link
DPCfam full seeds	DPCfam MC's seed sequence regions (SSRs) - based on the UniRef50 database (v. 2017_07)	Click me
DPCfam HMM profiles	HMM-profiles for all MCs	Click me
DPCfam MSA profiles	MSA-profiles for all MCs	Click me
AlphaFoldDB Reps	File containing details about representatives in AlphaFoldDB	Click me
DPCFamB	All files related to DPCFamB dataset	Click me

Case 2 : DPCStruct

The user can download the following files :

Table 11: DPCStruct files

File	Description	Link
DPCStruct Reps PDBs	DPCStruct MC's PDB files for each representative domain	Click me
DPCStruct Reps Seeds	DPCStruct MC's fasta files for each representative domain	Click me
DPCStruct MCs' Properties	CSV file containing properties of all MCs	Click me
DPCStruct vs PFam	CSV file containing infos related to the consistency of DPCStruct MCs with PFam	Click me
DPCStruct vs CATH	CSV file summarizing consistency with CATH	Click me
DPCStruct vs SCOPE	CSV file summarizing consistency with SCOPE	Click me

2.2.5 FAQs

The user can explore a list of frequently asked questions and their answers:

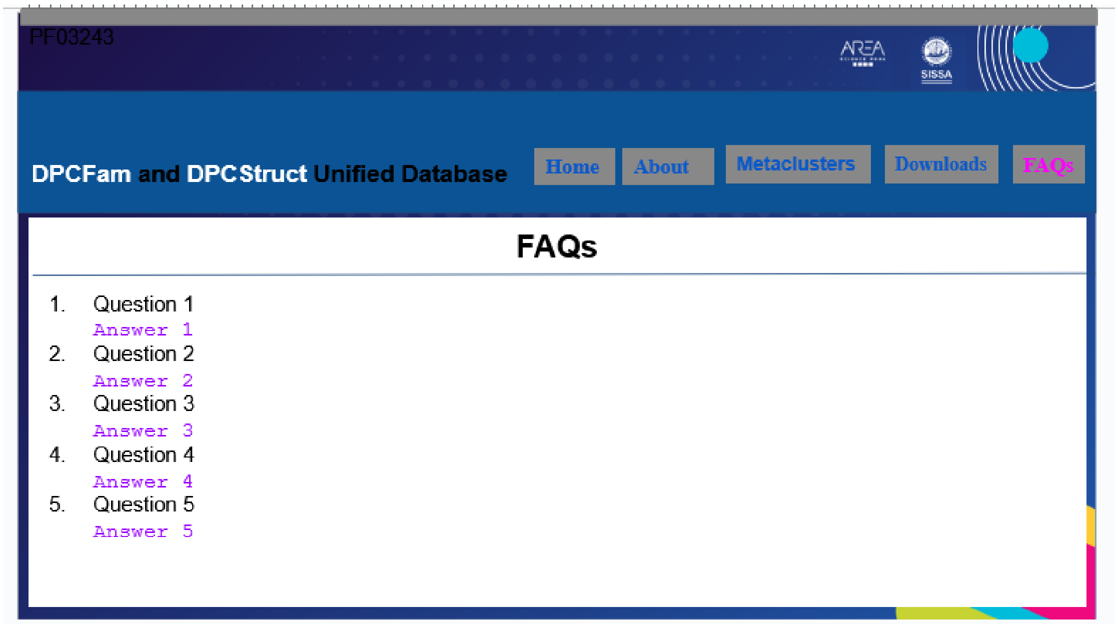


Figure 11: FAQs Page

3 Conclusion

Efforts are still being made to fully understand some relevant details and to improve this first draft as well. Likewise, the fact that our list of MC IDs (DPCFam) is not organised properly (3, 9, 10, 17, ...), even if it is sorted, could discourage a given user.

References

Barone, F., Laio, A., Punta, M., Cozzini, S., Ansuini, A., and Cazzaniga, A. (2025). Unsupervised domain classification of alphafold2-predicted protein structures. *PRX Life*, 3(2):023009.

Russo, E. T., Barone, F., Bateman, A., Cozzini, S., Punta, M., and Laio, A. (2022). Dpcfam: unsupervised protein family classification by density peak clustering of large sequence datasets. *PLOS Computational Biology*, 18(10):e1010610.